

A MICROSCOPIC DISCOVERY HAS INDIAN SCIENTISTS ON CLOUD NINE



BANGALORE | Signs of Life

It is a microscopic discovery but makes for a very big deal. Indian scientists have discovered three new alien species of ultra-violet resistant bacteria 40 km high in the sky, in the upper stratosphere. The species—*Janibacter hoylei*, *Bacillus isronensis* and *Bacillus aryabhata*—were detected in an experiment conducted using a 26.7-million-cubic-foot ISRO balloon carrying a 459-kg scientific payload soaked in 38 kg of liquid Neon, which was flown in from the National Balloon Facility in Hyderabad. Most space scientists who believe terrestrial life was brought down to earth by comets say the discovery of living samples as high as 40 km is significant because air cannot be transported above the local tropopause, which is at 16 km. The discovery should provide encouragement to those who study and reasearch the origins of life.

by Stephen David

250

sewage plants across the country are treating 25 per cent of the total sewage.

MYSORE | Spice Cadets

Indian food is on the verge of conquering the last frontier. The Defence Food Research Laboratory (DFRL) is in the middle of cooking up a formula to ensure that Indian astronauts of the future would be able to feed on the flavours of home without their digestive side-effects. The laboratory has worked around traditional Indian food like chicken curry, spinach, peas and mushroom to zero in on space food that is not spicy or high in fat content to ensure that, as the director of DFRL puts it, "the stomachs of the astronauts are not strained". All this so that when two Indian Air Force men are sent into orbit in 2015, they have no complaints about not being able to find good Indian food up there.

by Nirmala Ravindran

DELHI | To Each His Own

It was to be a meeting with Prime Minister Manmohan Singh to apprise him of India Inc's stand on the issue of unfair trade practices. The high profile turnout fell a little damp with the absence of Mukesh Ambani, who was reportedly out of the country. The tête-à-tête eventually turned out to be a one-stop shop for sectoral gains as corporates turned to their own agendas. While Rahul Bajaj brought up the issue of creeping acquisitions, Sunil Mittal wanted a boost to mobile broadband. Manmohan listened patiently and advised India Inc to be sensitive to those suffering most from the financial crisis. Some meltdown!

by Nivedita Mukherjee



(L-R) MANMOHAN WITH RATAN TATA AND SUNIL MITTAL

The Wasteland

India is dumping waste and dumping it big. The country's sewage and solid waste could grow at a compound annual growth rate of 2.5 per cent and 5 per cent respectively, according to a study by research and consulting firm Frost and Sullivan. By the end of 2009, the generation of sewage from urban centres alone could be 35,000 million litres per day while municipal solid waste generation could be 45 million tonnes. Presently, the 250-odd sewage treatment plants across the country are treating barely 25 per cent of the total sewage gen-

THE ISSUE OF URBAN WASTE MERITS ATTENTION



MANEESH AGNIHOTRI

erated. Blame it on shortage of funds, poor enforcement by pollution control boards, and poor awareness regarding protection of the environment, but India needs to sustain the massive growth that it is achieving, and without damaging the environment in the form of global warming, surface and ground water contamination. Experts believe that the cost of climate change in India could be a 9-13 per cent loss in GDP by 2100.

by Nandini Vaish